210217: metamonzogranite, South Well

(Goonetarra Granodiorite, Rothsay Suite, Annean Supersuite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Yalgoo (SH 50-2), Badja (2240)
MGA Zone 50, 478026E 6827893N

Sampled on 6 May 2013

This sample was collected from outcrop on Badja Station, about 4.6 km west-southwest of Moorgabby Well, 2.9 km east-southeast of Four Mile Well, and 2.5 km northwest of South Well.

Geological context

The unit sampled is the Goonetarra Granodiorite of the Rothsay Suite, Annean Supersuite (Van Kranendonk et al., 2013; Zibra, 2019). The 2763–2749 Ma Goonetarra Granodiorite consists mainly of metatonalite, metagranodiorite, and metamonzogranite, which are typically foliated and lineated (Zibra, 2019). In the Yalgoo dome, the Goonetarra Granodiorite is intruded into the Kynea Tonalite and into the greenstone succession (Zibra et al., 2018; Clos et al., 2018). Near the centre of the Yalgoo dome, the Goonetarra Granodiorite is intruded by monzogranite of the 2650–2600 Ma Walganna Suite of the Bald Rock Supersuite (Zibra, 2019). A monzogranite gneiss of the Goonetarra Granodiorite sampled about 10.6 km to the northeast yielded an igneous crystallization age of 2749 ± 4 Ma (GSWA 155822, Wingate et al., 2015). Another Goonetarra Granodiorite sample collected about 12.7 km to the south-southeast yielded an igneous crystallization age of 2752 ± 13 Ma (GSWA 155858, Lu et al., 2017).

Petrographic description

The sample is an inequigranular, coarse-grained metamonzogranite (Fig. 1), consisting of about 40% K-feldspar, 30% quartz, 25% plagioclase, 3–4% biotite, <1% magnetite, minor myrmekite, and accessory zircon. K-feldspar is microperthitic, anhedral to subhedral, roughly tabular, up to 12 mm in size, moderately strained, and exhibits minor microcline twinning. The larger grains contain abundant 0.1 to 1 mm long inclusions of plagioclase and rare quartz and biotite. Quartz is anhedral, irregular, up to 7 mm long, and some forms aggregates up to 9 mm in diameter of smaller quartz grains with serrated, intergrown boundaries. Quartz exhibits moderate undulose extinction and there are minor unstrained grains, up to 0.5 mm in size, along the boundaries of larger strained quartz grains. Plagioclase is anhedral to subhedral, equant to slightly tabular, and up to 5 mm in size. Minor myrmekite occurs at the boundaries between some plagioclase and K-feldspar. Biotite is mainly anhedral and irregular to subhedral, and up to 2 mm in size. Magnetite forms anhedral to subhedral, slightly elongate grains up to 1 mm in size and smaller irregular grains. Feldspars are very finely clay dusted and some plagioclase is weakly sericitized. Some biotite has been replaced by chlorite. The sample lacks a tectonic foliation but displays minor evidence of grain boundary adjustment. Microcline twinning and undulose extinction suggest the rock has been deformed and recovered at low to moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and subhedral to euhedral. The crystals are up to 450 µm long, and slightly elongate to acicular, with aspect ratios up to 10:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and most crystals contain high-uranium, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 30–31 July 2020, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.22% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seventeen analyses were obtained from 17 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Four analyses are >5% discordant or indicate high common Pb (*f*204 >1%). The dates obtained from these four analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 13 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Figure 1. Hand specimen image of sample 210217: metamonzogranite, South Well

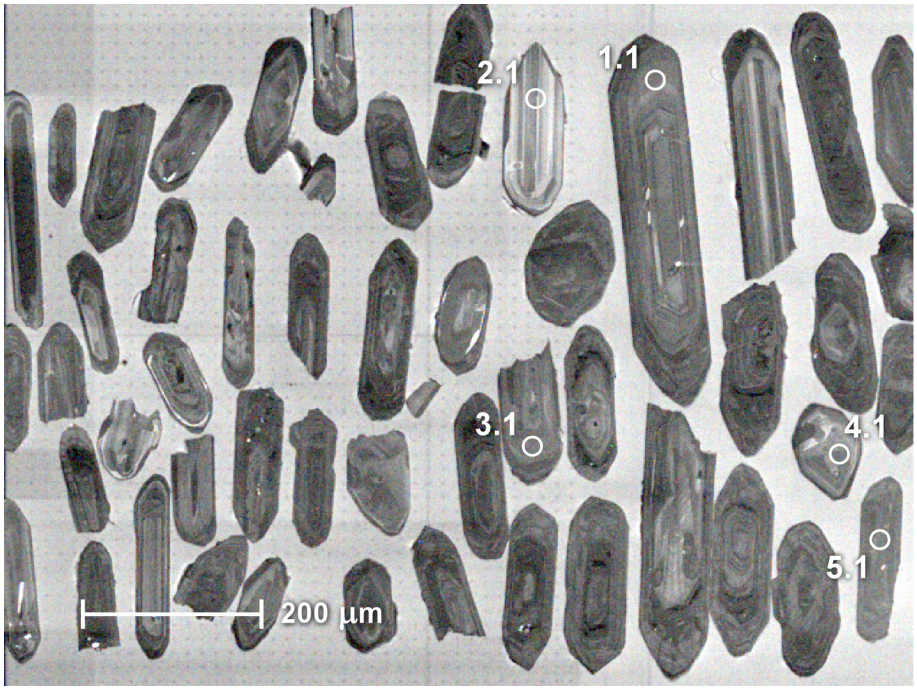

Figure 2. Cathodoluminescence image of representative zircons from sample 210217: metamonzogranite, South Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 210217: metamonzogranite, South Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	8	8.1	250	90	0.36	0.235	1.934	0.020	0.19160	0.00086	1.938	0.020	0.18950	0.00098	2682	23	2738	9	2.0
I	17	17.1	186	97	0.52	-0.064	1.942	0.022	0.18913	0.00101	1.940	0.022	0.18971	0.00105	2680	25	2740	9	2.2
I	6	6.1	284	112	0.39	0.137	1.922	0.019	0.19141	0.00132	1.924	0.019	0.19019	0.00136	2698	22	2744	12	1.7
I	4	4.1	228	127	0.56	0.444	1.939	0.020	0.19439	0.00093	1.948	0.021	0.19043	0.00121	2671	23	2746	10	2.7
I	3	3.1	447	51	0.11	0.046	1.925	0.017	0.19086	0.00061	1.926	0.017	0.19046	0.00063	2697	19	2746	5	1.8
I	14	14.1	314	38	0.12	0.043	1.967	0.019	0.19145	0.00075	1.968	0.019	0.19106	0.00077	2649	21	2751	7	3.7
I	1	1.1	312	81	0.26	0.024	1.873	0.018	0.19132	0.00073	1.874	0.018	0.19110	0.00074	2757	22	2752	6	-0.2
I	10	10.1	353	86	0.24	0.095	1.982	0.018	0.19210	0.00069	1.984	0.018	0.19125	0.00073	2631	20	2753	6	4.4
I	2	2.1	116	78	0.67	-0.066	1.972	0.025	0.19226	0.00120	1.971	0.025	0.19285	0.00125	2645	28	2767	11	4.4
I	7	7.1	203	100	0.49	-0.070	1.888	0.021	0.19269	0.00095	1.887	0.021	0.19331	0.00099	2742	25	2771	8	1.0
P	5	5.1	581	374	0.64	0.066	1.958	0.016	0.18910	0.00055	1.959	0.016	0.18852	0.00057	2659	18	2729	5	2.6
P	9	9.1	114	42	0.37	0.000	1.951	0.026	0.18877	0.00124	1.951	0.026	0.18877	0.00124	2668	29	2731	11	2.3
P	15	15.1	268	146	0.55	-0.050	1.936	0.019	0.18844	0.00080	1.935	0.019	0.18889	0.00083	2686	22	2733	7	1.7
D	13	13.1	173	146	0.84	4.962	1.751	0.067	0.23210	0.01409	1.843	0.079	0.18784	0.02302	2795	98	2723	202	-2.6
D	12	12.1	186	156	0.84	0.216	2.120	0.024	0.19224	0.00099	2.124	0.024	0.19031	0.00112	2487	23	2745	10	9.4
D	16	16.1	158	165	1.04	1.033	5.033	1.614	0.20192	0.00467	5.086	1.631	0.19271	0.00505	1157	340	2765	43	58.2
D	11	11.1	78	37	0.47	0.035	2.013	0.031	0.19399	0.00150	2.014	0.031	0.19368	0.00153	2599	33	2774	13	6.3

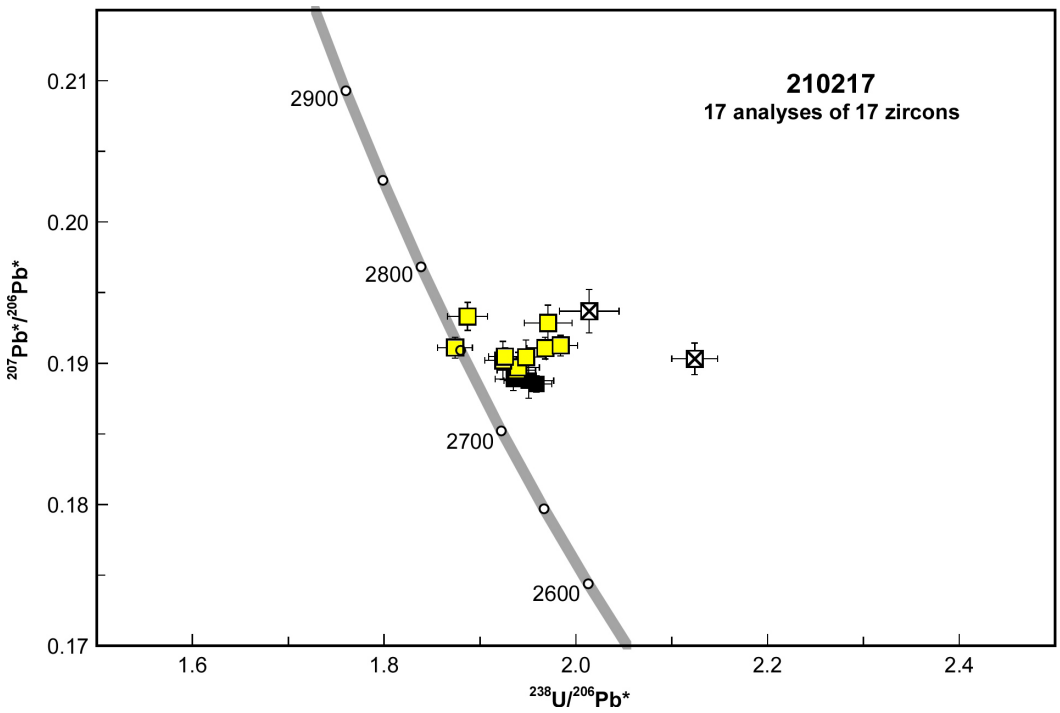

Figure 3. U–Pb analytical data for sample 210217: metamonzogranite, South Well. indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5% or high common Pb). Two analyses in Group D are not shown

Group I comprises 10 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2750 ± 7 Ma (MSWD = 1.4).

Group P comprises three analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2733–2729 Ma.

The date of 2750 ± 7 Ma for the 10 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite. The dates of 2733–2729 Ma for the three analyses in Group P are interpreted to reflect minor loss of radiogenic Pb.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Zibra, I 2021, 210217: metamonzogranite, South Well; Geochronology Record 1802: Geological Survey of Western Australia, 4p.

Data obtained: 31 July 2020

Data released: 21 October 2021